

BS-PH101 — Physics-I

Semester I/II • 3L:1T:0P • 4 Credits

Module 5 Comprehensive Exam Notes: Statistical Mechanics

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Macrostate, Microstate, Phase Space, and Ensembles

1.1 Basic Concepts of Statistical Mechanics

Statistical mechanics connects the microscopic behavior of individual atoms/molecules with the macroscopic thermodynamic properties (P, V, T, S) of a system.

Definitions: Macrostate, Microstate, and Phase Space

- **Macrostate:** The macroscopic state of a system specified by overall thermodynamic variables (e.g. (N, V, E) or (N, V, T)).
- **Microstate:** A specific detailed configuration of the system specifying the exact position ($\vec{r}_i$) and momentum ($\vec{p}_i$) of every individual particle.
- **Phase Space:** A $6N$ -dimensional conceptual space formed by 3 position coordinates (x, y, z) and 3 momentum coordinates (p_x, p_y, p_z) for each of N particles. The elementary phase cell volume in quantum mechanics is h^3 .
- **Thermodynamic Probability (W):** The total number of microstates corresponding to a given macrostate.

Boltzmann Entropy Relation

$$S = k_B \ln W \quad (1)$$

where $k_B = 1.38 \times 10^{-23}$ J/K is Boltzmann's constant, and W is the thermodynamic probability.

1.2 Statistical Ensembles

An **ensemble** is a large collection of independent, macroscopically identical systems existing in different microstates.

Three Main Gibbs Ensembles:

Ensemble	Fixed Parameters	Physical System Example
Microcanonical	Number N , Volume V , Energy E (Rigid, isolated, insulated walls).	Isolated system with zero heat exchange.
Canonical	Number N , Volume V , Temperature T (Diathermal walls in heat bath).	Closed system exchanging heat with a reservoir.
Grand Canonical	Chem. potential μ , Volume V , Temp. T (Permeable, diathermal walls).	Open system exchanging both energy and particles.

Solved Problem 1: Microstates and Entropy

Problem: Calculate the number of microstates and thermodynamic probability W for distributing $N = 4$ distinguishable particles in 2 equal compartments. Find the entropy of the most probable state.

Solution:

1. Total number of microstates = $2^N = 2^4 = 16$.
2. Macrostates are given by (n_1, n_2) where $n_1 + n_2 = 4$:
 - $(4, 0) : W = \frac{4!}{4!0!} = 1$
 - $(3, 1) : W = \frac{4!}{3!1!} = 4$
 - $(2, 2) : W = \frac{4!}{2!2!} = 6$ (Most Probable Macrostate)
 - $(1, 3) : W = 4$
 - $(0, 4) : W = 1$
3. Entropy of most probable macrostate $(2, 2)$:

$$S = k_B \ln(6) = (1.38 \times 10^{-23}) \times 1.7917 = 2.47 \times 10^{-23} \text{ J/K} \quad (2)$$

2. Density of States

2.1 Derivation of Density of States in 3D

The **Density of States** ($g(E)dE$) represents the number of available quantum states per unit volume in the energy range between E and $E + dE$.

3D Free Particle Density of States Derivation

In momentum space, momentum $p = \sqrt{2mE}$. The volume of momentum sphere of radius p is $V_p = \frac{4}{3}\pi p^3 = \frac{4}{3}\pi(2mE)^{3/2}$. The number of spatial-momentum phase cells of volume h^3 inside spatial volume V is:

$$N(E) = \frac{V \cdot \frac{4}{3}\pi(2mE)^{3/2}}{h^3} = \frac{4\pi V(2m)^{3/2}}{3h^3} E^{3/2} \quad (3)$$

Taking the derivative with respect to E gives $g(E) = \frac{dN(E)}{dE}$:

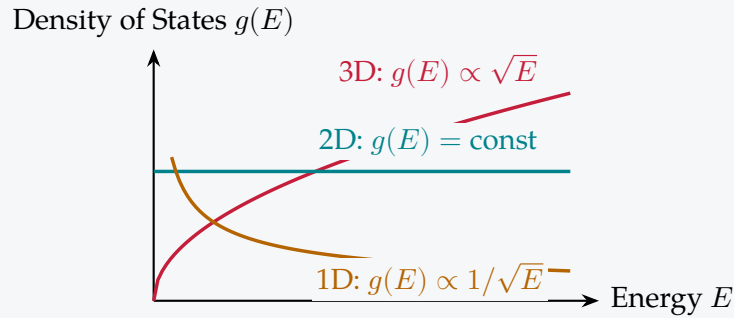
$$g(E)dE = \frac{4\pi V(2m)^{3/2}}{h^3} E^{1/2} dE \quad (4)$$

Including electron spin degeneracy $g_s = 2s + 1 = 2$ for electrons:

$$g_e(E)dE = \frac{8\pi V(2m)^{3/2}}{h^3} E^{1/2} dE \quad (5)$$

Density of States in Lower Dimensions:

1. **1D System:** $g_{1D}(E) \propto E^{-1/2}$.
2. **2D System:** $g_{2D}(E) = \text{constant}$ (independent of energy E).
3. **3D System:** $g_{3D}(E) \propto E^{1/2}$.

Density of States $g(E)$ in 1D, 2D, and 3D Systems

3. Maxwell-Boltzmann, Bose-Einstein, and Fermi-Dirac Statistics

3.1 Qualitative and Quantitative Treatment

Depending on particle distinguishability, quantum spin, and wave function symmetry:

Three Distribution Functions

- **Maxwell-Boltzmann (MB):** Classical, distinguishable particles, no spin restriction:

$$f_{\text{MB}}(E) = Ae^{-E/k_B T} \quad (6)$$

- **Bose-Einstein (BE):** Quantum, indistinguishable particles with integer spin ($s = 0, 1, 2, \dots$, Bosons):

$$f_{\text{BE}}(E) = \frac{1}{e^{(E-\mu)/k_B T} - 1} \quad (7)$$

- **Fermi-Dirac (FD):** Quantum, indistinguishable particles with half-integer spin ($s = 1/2, 3/2, \dots$, Fermions), obeys Pauli Exclusion Principle:

$$f_{\text{FD}}(E) = \frac{1}{e^{(E-E_F)/k_B T} + 1} \quad (8)$$

Comprehensive 8-Criterion Comparison Table:

Feature	Maxwell-Boltzmann	Bose-Einstein	Fermi-Dirac
Particle Nature	Classical, Distinguishable.	Quantum, Indistinguishable.	Quantum, Indistinguishable.
Spin	Any spin (Spin ignored).	Integral spin $(0, 1\hbar, 2\hbar)$.	Half-integral spin $(\frac{1}{2}\hbar, \frac{3}{2}\hbar)$.
Pauli Principle	Does not apply.	Does not apply.	Strictly obeys Pauli principle.
Occupancy/Cell	Unlimited $(0, 1, 2, \dots, \infty)$.	Unlimited $(0, 1, 2, \dots, \infty)$.	At most one particle (0 or 1).
Wave Function	No wave function overlap.	Symmetric under particle exchange.	Anti-symmetric under particle exchange.
Ground State	All particles rest at $E = 0$ as $T \rightarrow 0$.	Bose-Einstein Condensation at $T < T_c$.	Occupies up to Fermi level E_F at $T = 0$ K.
Physical Examples	Ideal gas molecules (He, Ar).	Photons, Phonons, α -particles, ^4He .	Free electrons, Protons, Neutrons, ^3He .

4. Fermi Energy and Bose-Einstein Condensation

4.1 Fermi Energy and Fermi-Dirac Function at $T = 0$ K

The **Fermi Energy** (E_F) is the highest filled energy level in a fermion system at absolute zero temperature ($T = 0$ K).

At $T = 0$ K:

$$f_{\text{FD}}(E) = \begin{cases} 1 & \text{if } E < E_F \\ 0 & \text{if } E > E_F \end{cases} \quad (9)$$

$$\text{Total number of free electrons } N = \int_0^{E_{F0}} g_e(E) dE = \int_0^{E_{F0}} \frac{8\pi V(2m)^{3/2}}{h^3} E^{1/2} dE = \frac{16\pi V(2m)^{3/2}}{3h^3} E_{F0}^{3/2}.$$

Fermi Energy and Average Energy at $T = 0$ K

Solving for E_{F0} in terms of electron density $n = N/V$:

$$E_{F0} = \frac{h^2}{8m} \left(\frac{3N}{\pi V} \right)^{2/3} = \frac{\hbar^2}{2m} (3\pi^2 n)^{2/3} \quad (10)$$

The total kinetic energy of N electrons at $T = 0$ K is $E_{\text{total}} = \int_0^{E_{F0}} E g(E) dE = \frac{3}{5} N E_{F0}$.
Average energy per electron at $T = 0$ K:

$$\langle E \rangle = \frac{3}{5} E_{F0} \quad (11)$$

Solved Problem 2: Fermi Energy of Copper

Problem: Copper has free electron density $n = 8.49 \times 10^{28} \text{ m}^{-3}$. Calculate: (a) Fermi energy E_F in eV, (b) Fermi velocity v_F .

Solution:

1. Fermi energy $E_{F0} = \frac{\hbar^2}{2m} (3\pi^2 n)^{2/3}$: $3\pi^2 n = 3\pi^2 (8.49 \times 10^{28}) = 2.514 \times 10^{29} \text{ m}^{-3}$.
 $(3\pi^2 n)^{2/3} = 3.983 \times 10^{19} \text{ m}^{-2}$.

$$E_{F0} = \frac{(1.054 \times 10^{-34})^2}{2 \times (9.11 \times 10^{-31})} \times 3.983 \times 10^{19} = 2.428 \times 10^{-18} \text{ J} = \mathbf{7.04 \text{ eV}} \quad (12)$$

2. Fermi velocity $v_F = \sqrt{\frac{2E_F}{m}} = \sqrt{\frac{2 \times 2.428 \times 10^{-18}}{9.11 \times 10^{-31}}} = \mathbf{1.57 \times 10^6 \text{ m/s}}$.

4.2 Bose-Einstein Condensation (BEC)

At extremely low temperatures below a critical transition temperature T_c , a substantial fraction of bosons collapse into the lowest quantum ground state ($E = 0$), forming a macroscopic quantum state known as **Bose-Einstein Condensate**.

Critical Temperature and Ground State Fraction

Critical BEC transition temperature T_c :

$$T_c = \frac{\hbar^2}{2\pi m k_B} \left(\frac{N}{2.612 V} \right)^{2/3} \quad (13)$$

Fraction of bosons in ground state for $T < T_c$:

$$\frac{N_0}{N} = 1 - \left(\frac{T}{T_c} \right)^{3/2} \quad (14)$$

Quick Revision — Module 5

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Key Formulas and Metrics at a Glance

Boltzmann Entropy Relation	$S = k_B \ln W$ (W = thermodynamic probability)
Elementary Phase Cell	$d\Omega = h^3$ (quantum phase space volume)
3D Density of States	$g_e(E)dE = \frac{8\pi V(2m)^{3/2}}{h^3} E^{1/2} dE$
MB Distribution Function	$f_{\text{MB}}(E) = Ae^{-E/k_B T}$
BE Distribution Function	$f_{\text{BE}}(E) = \frac{1}{e^{(E-\mu)/k_B T} - 1}$ (Bosons, Integer spin)
FD Distribution Function	$f_{\text{FD}}(E) = \frac{1}{e^{(E-E_F)/k_B T} + 1}$ (Fermions, Half-spin)
Fermi Energy ($T = 0$ K)	$E_{F0} = \frac{h^2}{8m} \left(\frac{3N}{\pi V} \right)^{2/3} = \frac{h^2}{2m} (3\pi^2 n)^{2/3}$
Average Fermi Energy	$\langle E \rangle = \frac{3}{5} E_{F0}$ (at $T = 0$ K)
BEC Transition Temp.	$T_c = \frac{h^2}{2\pi m k_B} \left(\frac{N}{2.612 V} \right)^{2/3}$
BEC Condensate Fraction	$\frac{N_0}{N} = 1 - \left(\frac{T}{T_c} \right)^{3/2}$ (for $T < T_c$)

One-Line Definitions

- **Microstate:** A specific detailed microscopic specification of position and momentum coordinates for all particles in a system.
- **Ensemble:** A large virtual collection of independent, macroscopically identical systems in different microstates.
- **Fermi Energy (E_F):** The maximum energy state occupied by free electrons at absolute zero temperature ($T = 0$ K).
- **Boson:** A particle with integral quantum spin ($0, 1\hbar, 2\hbar$) obeying Bose-Einstein statistics and symmetric under particle exchange.
- **Bose-Einstein Condensation:** A phase transition where a macroscopic number of bosons collapse into the single lowest energy ground state at $T < T_c$.

Frequently Asked / Viva Questions

Q1. What is the difference between a macrostate and a microstate?

A macrostate specifies overall macroscopic variables (N, V, E); a microstate specifies exact microscopic position and momentum coordinates ($\vec{r}_i, \vec{p}_i$) of every particle.

Q2. What is phase space and what is its minimum volume in quantum mechanics?

Phase space is a $6N$ -dimensional space combining 3 position and 3 momentum

Frequently Asked / Viva Questions (contd.)

axes per particle. Its minimum volume cell in quantum mechanics is h^3 .

Q3. Distinguish between Microcanonical, Canonical, and Grand Canonical ensembles.

Microcanonical keeps (N, V, E) fixed; Canonical keeps (N, V, T) fixed (heat exchange allowed); Grand Canonical keeps (μ, V, T) fixed (heat and particle exchange allowed).

Q4. State the basic difference between Bosons and Fermions.

Bosons have integral spin $(0, 1\hbar)$, symmetric wavefunctions, and do not obey Pauli principle. Fermions have half-integral spin $(\frac{1}{2}\hbar)$, anti-symmetric wavefunctions, and strictly obey Pauli principle.

Q5. Why does density of states in 3D vary as $\sqrt{E}$?

Because the momentum sphere volume $V_p \propto p^3 \propto E^{3/2}$. Taking the derivative $g(E) = dN/dE$ gives $g(E) \propto E^{1/2} = \sqrt{E}$.

Q6. Draw Fermi-Dirac distribution curve at $T = 0$ K and $T > 0$ K.

At $T = 0$ K, $f(E)$ is a step function ($f = 1$ for $E < E_F$, $f = 0$ for $E > E_F$). At $T > 0$ K, thermal agitation creates a symmetric tail around E_F where $f(E_F) = 1/2$.

Q7. What is the physical meaning of Fermi energy?

Fermi energy represents the highest energy level occupied by free conduction electrons in a metal at absolute zero temperature ($T = 0$ K).

Q8. Calculate the average energy of an electron gas at $T = 0$ K in terms of E_F .

$$\langle E \rangle = \frac{\int_0^{E_F} E \cdot E^{1/2} dE}{\int_0^{E_F} E^{1/2} dE} = \frac{\frac{2}{5} E_F^{5/2}}{\frac{2}{3} E_F^{3/2}} = \frac{3}{5} E_F.$$

Q9. Under what conditions do BE and FD statistics reduce to MB statistics?

At high temperatures ($T \rightarrow \infty$) and low particle densities ($n \rightarrow 0$), when $e^{(E-\mu)/k_B T} \gg 1$, both BE and FD distributions reduce to classical MB distribution $f(E) \approx e^{-(E-\mu)/k_B T}$.

Q10. What is Bose-Einstein Condensation (BEC)?

BEC is a low-temperature phenomenon where a macroscopic fraction of bosons condense into the single zero-momentum ground state at $T < T_c$.

Q11. Why can photons undergo BEC without fixing total particle number N ?

Photons have chemical potential $\mu = 0$ because photon number is not conserved (photons are freely created/absorbed by cavity walls).

Q12. Why do electrons in a metal not contribute significantly to heat capacity at room temperature?

Because Pauli exclusion principle restricts thermal excitation ($k_B T$) to only a small fraction ($\sim k_B T / E_F \approx 1\%$) of electrons near the Fermi level E_F .

Summary Comparison of Quantum Statistics

Property	Maxwell-Boltzmann	Bose-Einstein	Fermi-Dirac
Distribution Eq.	$f(E) \propto e^{-E/k_B T}$	$f(E) = \frac{1}{e^{(E-\mu)/k_B T} - 1}$	$f(E) = \frac{1}{e^{(E-E_F)/k_B T} + 1}$
Ground State Occupancy	Non-zero classical energy.	Condensate (N_0 particles at $E = 0$).	Filled Fermi sphere up to E_F .
Wave Symmetry	Distinguishable (No symmetry).	Symmetric under exchange.	Anti-symmetric under exchange.